

Lecture Notes on Complex Numbers

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Chapter 1

Complex Numbers

1.1 Sums and Products

The complex numbers can be defined as ordered pairs of real numbers (x, y) subject to specific operations of addition and multiplication. We identify the set of complex numbers $\mathbb{C}$ with the xy -plane (the complex plane).

Definition 1.1.1 (Complex Number). A complex number z is defined as an ordered pair of real numbers:

$$z = (x, y)$$

where x is the *real part* of z , denoted $\operatorname{Re}(z)$, and y is the *imaginary part* of z , denoted $\operatorname{Im}(z)$.

Two complex numbers $z_1 = (x_1, y_1)$ and $z_2 = (x_2, y_2)$ are equal if and only if their real parts are equal and their imaginary parts are equal.

Definition 1.1.2 (Algebraic Operations). Let $z_1 = (x_1, y_1)$ and $z_2 = (x_2, y_2)$. We define sum and product as:

$$\begin{aligned} z_1 + z_2 &= (x_1 + x_2, y_1 + y_2) \\ z_1 z_2 &= (x_1 x_2 - y_1 y_2, y_1 x_2 + x_1 y_2) \end{aligned}$$

Remark 1.1.1. The set of complex numbers contains the set of real numbers as a subset. We identify the real number x with the pair $(x, 0)$. Under this identification, operations correspond to standard real arithmetic.

1.1.1 The Imaginary Unit

It is customary to denote the number $(0, 1)$ by i . Using the definition of multiplication:

$$i^2 = (0, 1)(0, 1) = (0 \cdot 0 - 1 \cdot 1, 1 \cdot 0 + 0 \cdot 1) = (-1, 0) = -1$$

Thus, we can write any complex number $z = (x, y)$ as:

$$z = x + iy$$

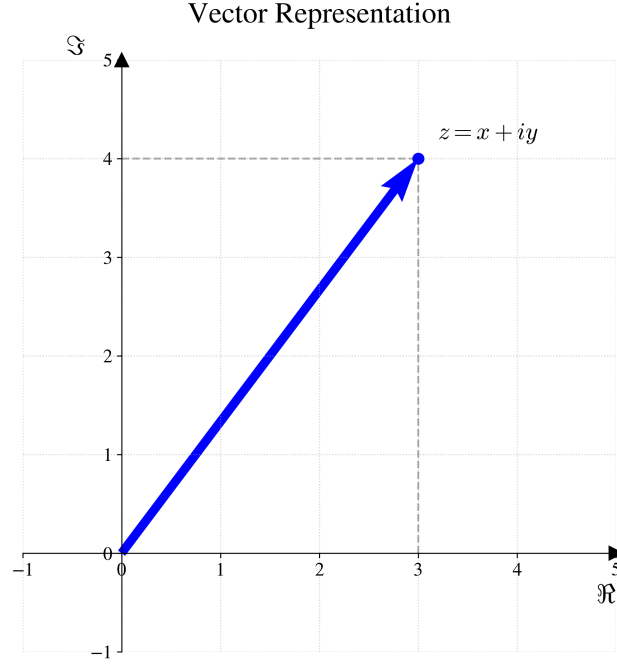


Figure 1.1: Vector representation of a complex number $z = x + iy$. The point (x, y) corresponds to the complex number in the plane.

1.2 Basic Algebraic Properties

The set $\mathbb{C}$ forms a **field** under addition and multiplication.

Theorem 1.2.1 (Field Properties). *For any $z_1, z_2, z_3 \in \mathbb{C}$, the following laws hold:*

1. **Commutative laws:** $z_1 + z_2 = z_2 + z_1$ and $z_1 z_2 = z_2 z_1$.
2. **Associative laws:** $(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$ and $(z_1 z_2) z_3 = z_1 (z_2 z_3)$.
3. **Distributive law:** $z(z_1 + z_2) = z z_1 + z z_2$.
4. **Identities:** Additive identity is $0 = (0, 0)$. Multiplicative identity is $1 = (1, 0)$.
5. **Inverses:** For every z , there is an additive inverse $-z$. For every $z \neq 0$, there is a multiplicative inverse z^{-1} .

Example 1.2.1. Find the multiplicative inverse of $z = x + iy$ assuming $z \neq 0$. We seek $u + iv$ such that $(x + iy)(u + iv) = 1$. Solving the system of linear equations yields:

$$z^{-1} = \left(\frac{x}{x^2 + y^2}, \frac{-y}{x^2 + y^2} \right)$$

1.3 Vectors and Moduli

A complex number $z = x + iy$ is naturally associated with a position vector in the plane starting from the origin and terminating at (x, y) .

Definition 1.3.1 (Modulus). The modulus (or absolute value) of $z = x + iy$ is the non-negative real number:

$$|z| = \sqrt{x^2 + y^2}$$

Geometrically, this represents the distance of the point z from the origin.

Lemma 1.3.1 (Distance between points). *The distance between two complex numbers z_1 and z_2 is given by $|z_1 - z_2|$.*

Exercise 1.3.1. Prove that $|z|^2 = (\operatorname{Re}(z))^2 + (\operatorname{Im}(z))^2$.

1.4 The Triangle Inequality

One of the most crucial inequalities in analysis is the Triangle Inequality, which states that the length of a side of a triangle is less than or equal to the sum of the lengths of the other two sides.

Theorem 1.4.1 (Triangle Inequality). *For any $z_1, z_2 \in \mathbb{C}$:*

$$|z_1 + z_2| \leq |z_1| + |z_2|$$

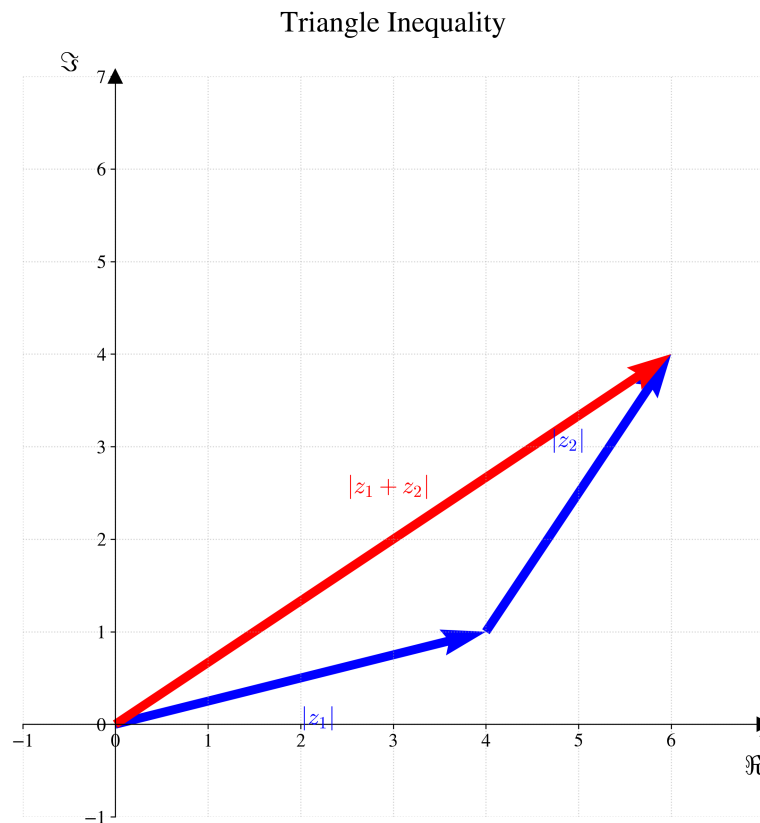


Figure 1.2: Geometric representation of the Triangle Inequality for complex numbers z_1 and z_2 . The length of the vector $z_1 + z_2$ is less than or equal to the sum of the lengths of z_1 and z_2 .

Corollary 1.4.2. *It follows from the theorem that:*

$$|z_1 + z_2| \geq ||z_1| - |z_2||$$

See Figure 1.2 for a geometric illustration.

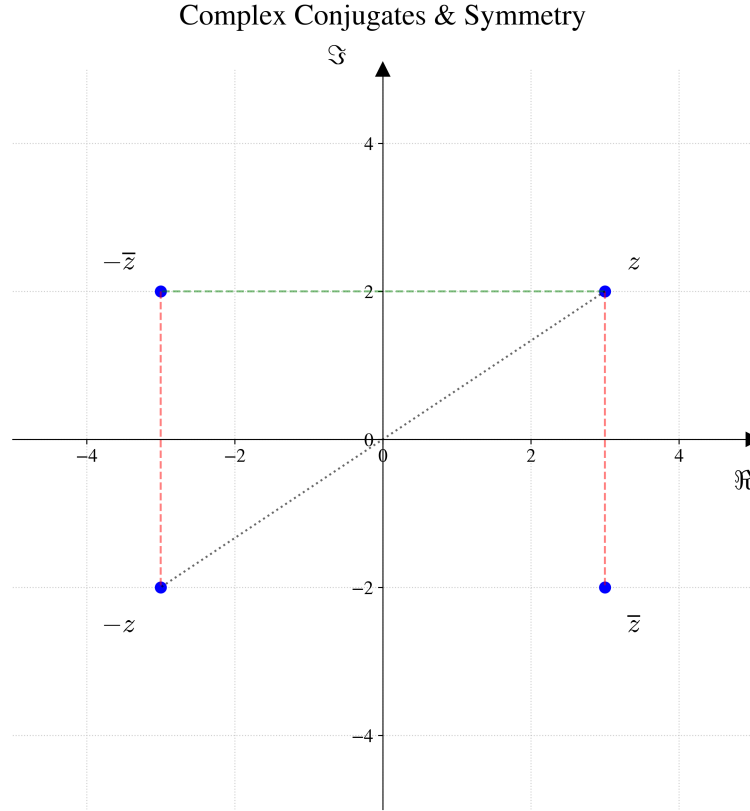


Figure 1.3: Geometric representation of the complex conjugate. The point $z = x + iy$ is reflected across the real axis to obtain $\bar{z} = x - iy$; the dashed line indicates the reflection. Analogously, $-z$ is the reflection across the origin, and $-\bar{z}$ is the reflection across the imaginary axis.

1.5 Complex Conjugates

Definition 1.5.1 (Conjugate). The complex conjugate of a number $z = x + iy$ is defined as:

$$\bar{z} = x - iy$$

Geometrically, $\bar{z}$ is the reflection of the point z across the real axis.

Theorem 1.5.1 (Properties of Conjugates). *For all $z, z_1, z_2 \in \mathbb{C}$:*

1. $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$
2. $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$
3. $|z|^2 = z\bar{z}$
4. $\operatorname{Re}(z) = \frac{z + \bar{z}}{2}$ and $\operatorname{Im}(z) = \frac{z - \bar{z}}{2i}$

1.6 Exponential Form

Using polar coordinates (r, θ) where $x = r \cos \theta$ and $y = r \sin \theta$, we can express complex numbers in exponential form.

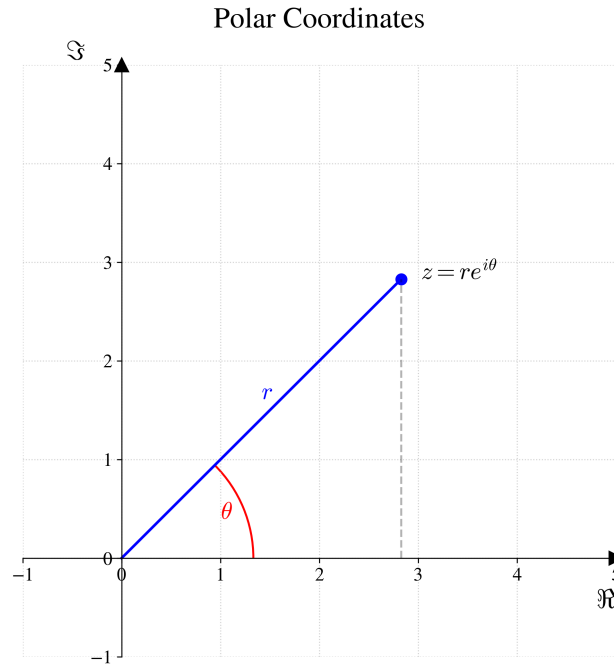


Figure 1.4: Exponential form of a complex number $z = re^{i\theta}$. The modulus r represents the distance from the origin, and the angle θ (argument) represents the direction from the positive real axis.

Definition 1.6.1 (Argument). The argument of z , denoted $\arg(z)$, is the set of angles θ such that $z = r(\cos \theta + i \sin \theta)$.

$$\theta = \arg(z) = \Theta + 2n\pi, \quad n \in \mathbb{Z}$$

where Θ is the **Principal Argument**, denoted $\text{Arg}(z)$, such that $-\pi < \Theta \leq \pi$.

Theorem 1.6.1 (Euler's Formula). For any real number θ ,

$$e^{i\theta} = \cos \theta + i \sin \theta$$

Thus, any non-zero complex number can be written as:

$$z = re^{i\theta}$$

1.7 Products and Powers in Exponential Form

Exponential form makes multiplication and division significantly easier geometrically.

Theorem 1.7.1 (Multiplication in Polar Form). If $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$, then:

$$z_1 z_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)}$$

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}$$

Remark 1.7.1. Multiplication by z scales the modulus by $|z|$ and rotates the vector by $\arg(z)$.

1.7.1 De Moivre's Formula

From the multiplication rule, we derive the formula for integer powers.

Theorem 1.7.2 (De Moivre's Formula). *For any integer n ,*

$$(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta)$$

or equivalently, $(e^{i\theta})^n = e^{in\theta}$.

Example 1.7.1. Calculate $(1 + i)^{10}$. First, convert $1 + i$ to exponential form:

$$r = \sqrt{1^2 + 1^2} = \sqrt{2}, \quad \text{Arg}(z) = \frac{\pi}{4}$$

$$z = \sqrt{2}e^{i\pi/4}$$

$$z^{10} = (\sqrt{2})^{10}e^{i10\pi/4} = 2^5e^{i5\pi/2} = 32e^{i(\pi/2+2\pi)} = 32e^{i\pi/2} = 32i$$

1.8 Roots of Complex Numbers

Finding the n -th roots of a complex number z_0 involves solving $z^n = z_0$.

Theorem 1.8.1 (Roots). *The n distinct n -th roots of $z_0 = r_0e^{i\theta_0}$ are given by:*

$$z_k = \sqrt[n]{r_0} \exp\left(i \frac{\theta_0 + 2k\pi}{n}\right), \quad k = 0, 1, \dots, n-1$$

Geometrically, the roots lie on a circle of radius $\sqrt[n]{r_0}$ centered at the origin and form the vertices of a regular n -sided polygon.

1.9 Regions in the Complex Plane

To do calculus, we need topological definitions regarding sets of points in the plane.

Definition 1.9.1 (Epsilon Neighborhood). An ε -neighborhood of a point z_0 is the set of all points z such that:

$$|z - z_0| < \varepsilon$$

Definition 1.9.2 (Topological Concepts). The following definitions apply to sets of complex numbers:

- **Interior Point:** A point z_0 in a set S is an interior point if there exists some neighborhood of z_0 completely contained in S .
- **Open Set:** A set S is open if every point in S is an interior point.
- **Closed Set:** A set S is closed if it contains all its boundary points.
- **Domain:** An open, connected set.
- **Region:** A domain together with some, none, or all of its boundary points.

Exercise 1.9.1. Determine if the set $S = \{z \in \mathbb{C} \mid |z| < 1\}$ is open or closed.

Solution: It is open, since for each point z_0 in S , we can find define $\varepsilon = \frac{1}{3}\text{dist}(z_0, \partial S)$; then the set $\mathcal{U}_\varepsilon(z_0) = \{z \in \mathbb{C} \mid |z - z_0| < \varepsilon\}$ is an ε -neighborhood that lies entirely within S .

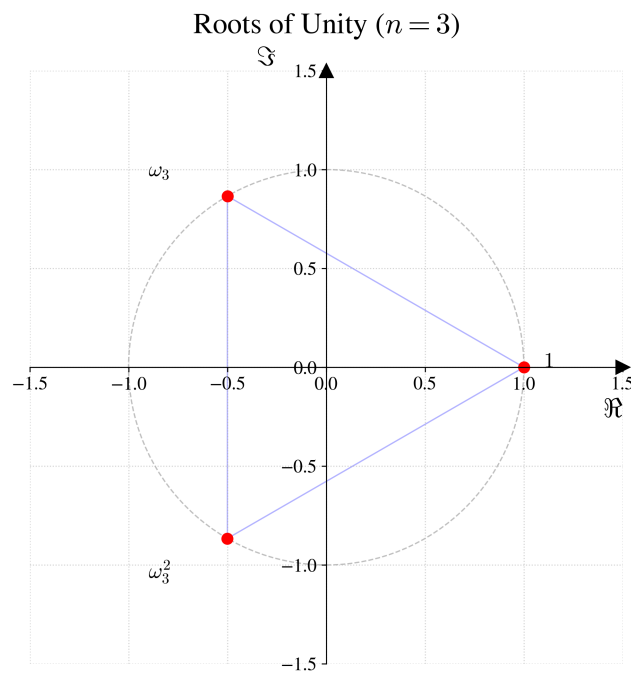


Figure 1.5: The n -th roots of a complex number z_0 are evenly spaced on a circle in the complex plane, forming the vertices of a regular n -sided polygon.

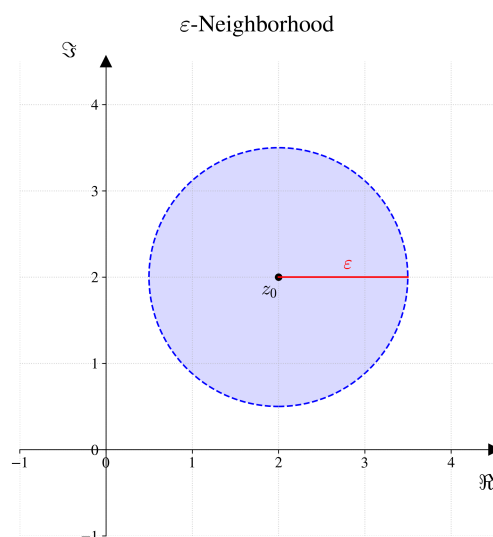


Figure 1.6: An ε -neighborhood around the point z_0 . The shaded area represents all points z such that $|z - z_0| < \varepsilon$.

1.10 Exercises

Exercise 1.10.1. Write the following complex numbers in the form $x + iy$:

1. $(5 - 2i) + (3 + 4i)$
2. $(2 + 3i)(4 - i)$
3. $\frac{1}{1+i}$
4. $\frac{3-2i}{-1+i}$

Exercise 1.10.2. Find $\operatorname{Re}(z)$ and $\operatorname{Im}(z)$ for the following:

1. $z = \frac{1}{i}$
2. $z = (1 - i)^3$
3. $z = \frac{2+i}{3-4i} + \frac{2-i}{5i}$

Exercise 1.10.3. Let $z_1 = 1 - i$ and $z_2 = -2 + 4i$. Compute:

1. $z_1^2 - 2z_1 + 3$
2. $|2z_1 - 3z_2|$
3. $\operatorname{Re}(z_1 z_2)$

Exercise 1.10.4. Verify that $(\sqrt{2} - i) - i(1 - \sqrt{2}i) = -2i$.

Exercise 1.10.5. Solve the following equation for z :

$$z(1 + i) = \bar{z} + (3 + 2i)$$

Hint: Let $z = x + iy$ and equate real and imaginary parts.

Exercise 1.10.6. Verify the following properties using $z = x + iy$:

1. $i\bar{z} = -i\bar{z}$
2. $\overline{\bar{z} + 3i} = z - 3i$
3. $|z|^2 = z\bar{z}$

Exercise 1.10.7. Prove that for any $z_1, z_2 \in \mathbb{C}$:

$$|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2(|z_1|^2 + |z_2|^2)$$

This is known as the *Parallelogram Law*.

Exercise 1.10.8. Show that $|z_1 - z_2|^2 = |z_1|^2 + |z_2|^2 - 2\operatorname{Re}(z_1 \bar{z}_2)$.

Exercise 1.10.9. (*Hard*) If $|z| = 1$, prove that:

$$\left| \frac{z - w}{1 - \bar{w}z} \right| = 1$$

for any complex number w with $|w| \neq 1$.

Exercise 1.10.10. Write the following numbers in exponential form $re^{i\theta}$ ($-\pi < \theta \leq \pi$):

1. $1 - i$
2. $-2\sqrt{3} - 2i$
3. $-i$
4. $(1 - i\sqrt{3})^2$

Exercise 1.10.11. Find the Principal Argument, $\text{Arg}(z)$, for:

1. $z = \frac{-2}{1+i\sqrt{3}}$
2. $z = (\sqrt{3} - i)^6$

Exercise 1.10.12. Using exponential form, show that:

$$(-1 + i)^7 = -8(1 + i)$$

Exercise 1.10.13. Compute the following powers and write the result in the form $x + iy$:

1. $(1 + i)^{10}$
2. $\left(\frac{1-i\sqrt{3}}{2}\right)^{15}$
3. $(1 - i)^{-8}$

Exercise 1.10.14. Compute the value of $(1 + \cos \frac{\pi}{3} + i \sin \frac{\pi}{3})^6$.

Hint: Use De Moivre's formula.

Exercise 1.10.15. Find all distinct roots of the following equations and locate them in the complex plane:

1. $z^3 = 1$
2. $z^2 + 2i = 0$
3. $z^4 = -16$

Exercise 1.10.16. Find the four values of $(-1)^{1/4}$.

Exercise 1.10.17. Solve the quadratic equation $z^2 + z + 1 = 0$ for z .

Exercise 1.10.18. Sketch the set of points determined by the condition:

1. $|z - 1 + i| = 1$
2. $|z + i| \leq 3$
3. $\text{Re}(\bar{z} - i) = 2$
4. $|2z - i| = 4$

Exercise 1.10.19. Describe the set of points z such that $\text{Im}(z^2) > 0$.

Exercise 1.10.20. Prove that if $1 + z + z^2 + \cdots + z^n = 0$, then z is a root of unity, i.e. $z = e^{\frac{2\pi ik}{n+1}}$ for some integer k .

Exercise 1.10.21. Establish the Lagrange trigonometric identity:

$$1 + \cos \theta + \cos 2\theta + \cdots + \cos n\theta = \frac{1}{2} + \frac{\sin[(n + 1/2)\theta]}{2 \sin(\theta/2)}$$

where $0 < \theta < 2\pi$.

Chapter 2

Analytic Functions

2.1 Functions of a Complex Variable

A complex function f is a rule that assigns to each complex number z in a set $S \subseteq \mathbb{C}$ a complex number w . We write:

$$w = f(z)$$

Here, S is the **domain of definition** of f . Since $z = x + iy$, the function $f(z)$ can be decomposed into its real and imaginary parts:

$$f(z) = u(x, y) + iv(x, y)$$

where u and v are real-valued functions of the real variables x and y .

Example 2.1.1. Consider $f(z) = z^2$.

$$f(x + iy) = (x + iy)^2 = x^2 - y^2 + i(2xy)$$

Thus, $u(x, y) = x^2 - y^2$ and $v(x, y) = 2xy$.

We cannot always draw complex functions, actually we usually cannot draw them at all since they map from a 2D space (the complex plane) to another 2D space (another complex plane), resulting in a 4D space which is not directly visualizable. However, we can analyze how certain sets in the z -plane are transformed into sets in the w -plane. This is often done by examining the images of lines, circles, and other simple shapes under the mapping defined by f . For this reason, we usually think about complex function as mappings from the z -plane to the w -plane. Therefore, we describe this function by means of their **geometric** properties. For example

1. We say that the map $w = z + 1 + i$ is a translation of the z -plane by the vector $(1, 1)$.
2. The map $w = \bar{z}$ is a reflection about the real axis.
3. The map $w = iz$ is a rotation of the z -plane by 90° counterclockwise about the origin.

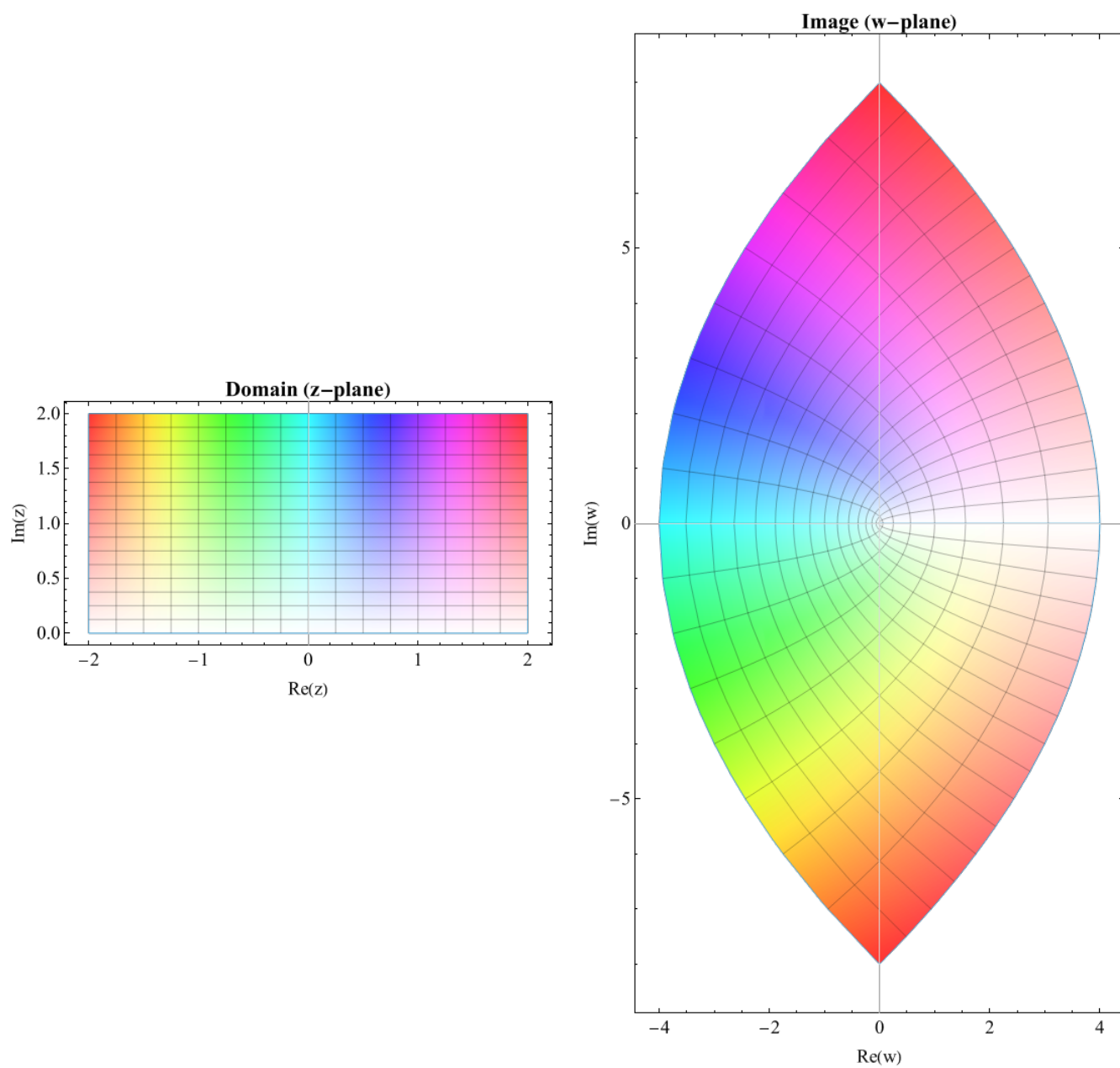
Mapping $w = z^2$ 

Figure 2.1: Mapping of a complex function from the z -plane to the w -plane.

2.1.1 Real valued functions

A special case of complex functions are those that map complex numbers to real numbers, i.e., $f : \mathbb{C} \rightarrow \mathbb{R}$. In this case, the imaginary part $v(x, y)$ is identically zero, this complex functions becomes identical to a real-valued function of two real variables:

$$f(z) = u(x, y) + i \cdot 0 = u(x, y),$$

therefore we can draw their graphs in the usual 3D space with axes x, y, u . Another way of getting information from complex valued functions is to look at the graph of their real and imaginary parts separately as surfaces in 3D space.

2.1.2 Example: The Joukowski Transformation

Let us examine the mapping defined by the function:

$$w = f(z) = z + \frac{1}{z}$$

This function is historically significant in aerodynamics for mapping circles into airfoil shapes. To understand the geometry of this mapping, it is most convenient to use polar coordinates $z = re^{i\theta}$. Substituting into the function:

$$w = re^{i\theta} + \frac{1}{re^{i\theta}} = r(\cos \theta + i \sin \theta) + \frac{1}{r}(\cos \theta - i \sin \theta)$$

Separating the real and imaginary parts $w = u + iv$, we obtain:

$$u = \left(r + \frac{1}{r}\right) \cos \theta, \quad v = \left(r - \frac{1}{r}\right) \sin \theta$$

Mapping of the Unit Circle ($r = 1$)

Consider the unit circle $|z| = 1$. Here $r = 1$, so the equations become:

$$u = 2 \cos \theta, \quad v = 0$$

As θ varies from 0 to 2π , u varies from 2 to -2 and back to 2. Thus, the unit circle in the z -plane is mapped onto the line segment $[-2, 2]$ on the real axis of the w -plane.

Mapping of Circles ($r = c > 1$)

Now consider a circle $|z| = c$ where $c > 1$. The equations for u and v are:

$$u = A \cos \theta, \quad v = B \sin \theta$$

where $A = c + \frac{1}{c}$ and $B = c - \frac{1}{c}$. Using the identity $\cos^2 \theta + \sin^2 \theta = 1$, we can eliminate θ :

$$\frac{u^2}{A^2} + \frac{v^2}{B^2} = 1$$

This is the equation of an ellipse with semi-major axis A and semi-minor axis B . Note that the foci of this ellipse are located at $\pm\sqrt{A^2 - B^2} = \pm\sqrt{4} = \pm 2$, which are the endpoints of the slit created by the unit circle mapping.

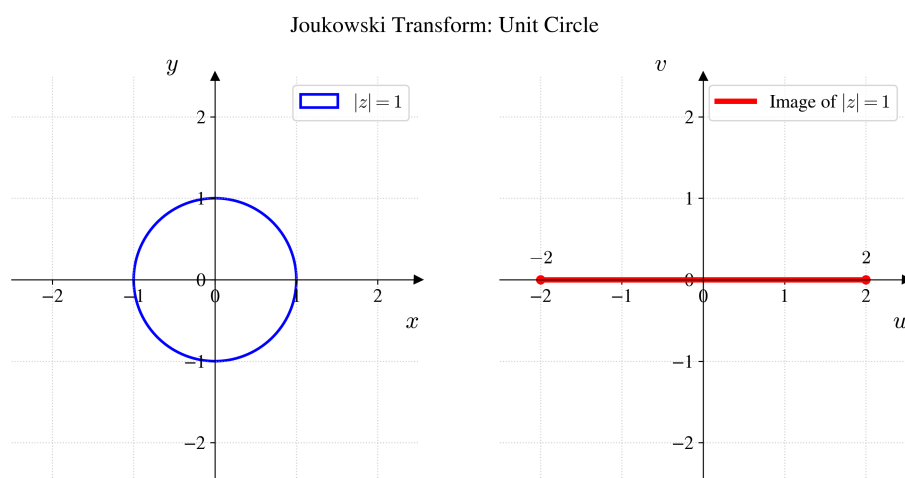
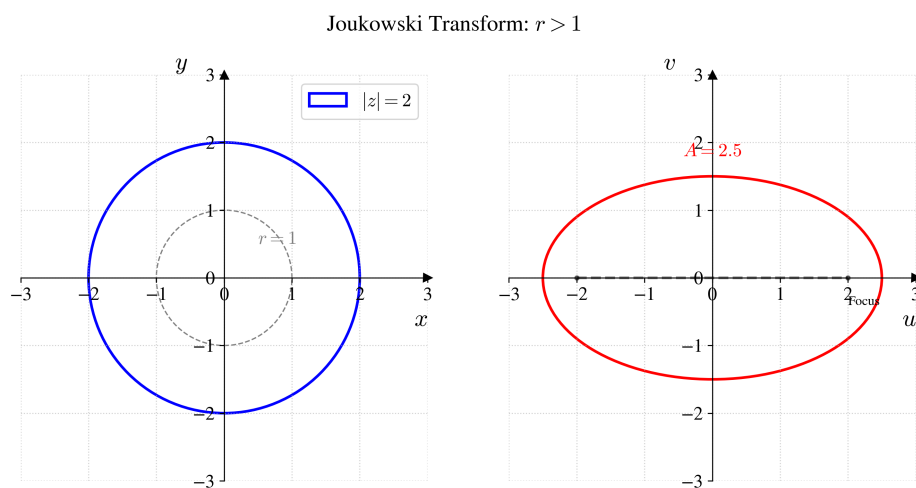


Figure 2.2: Mapping of the unit circle under the Joukowski transformation.

Figure 2.3: Mapping of the circle $|z| = c$ under the Joukowski transformation.

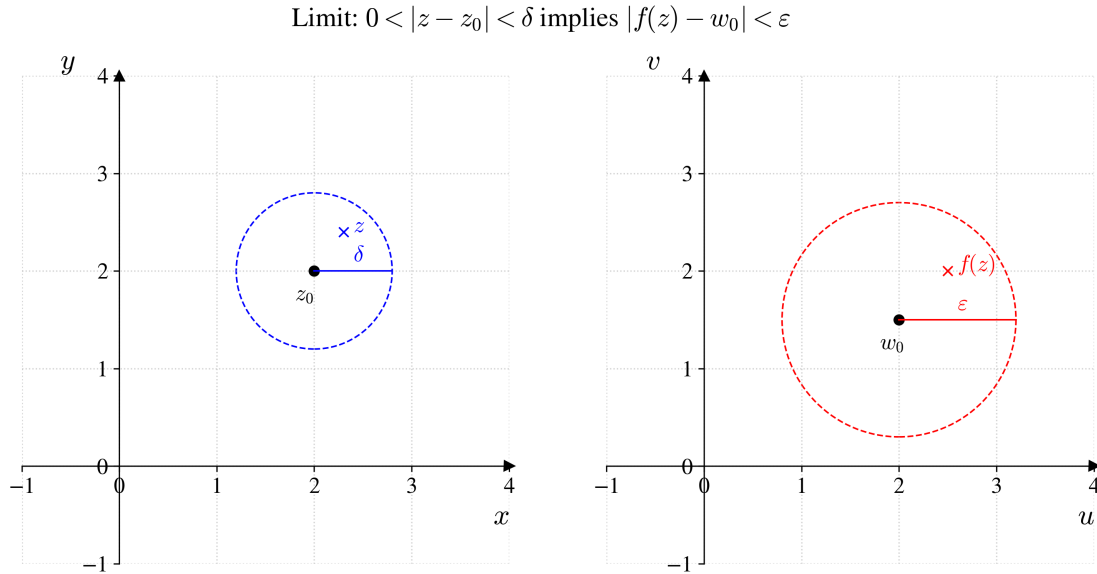


Figure 2.4: Illustration of the limit definition in the complex plane.

2.2 Limits and Continuity

The concept of a limit in the complex plane is similar to that in real calculus, but since the neighborhood is two-dimensional, z can approach z_0 from any direction.

Definition 2.2.1 (Limit). Let f be defined in a neighborhood of z_0 , except possibly at z_0 itself. We say that the limit of $f(z)$ as z approaches z_0 is w_0 , written

$$\lim_{z \rightarrow z_0} f(z) = w_0$$

if, for every $\varepsilon > 0$, there exists a $\delta > 0$ such that:

$$|f(z) - w_0| < \varepsilon \quad \text{whenever} \quad 0 < |z - z_0| < \delta$$

Theorem 2.2.1 (Continuity). A function f is continuous at z_0 if:

1. $f(z_0)$ is defined,
2. $\lim_{z \rightarrow z_0} f(z)$ exists, and
3. $\lim_{z \rightarrow z_0} f(z) = f(z_0)$.

2.3 Derivatives

Differentiation is the cornerstone of complex analysis. The definition is formally identical to the real case, but the implications are far more profound.

Definition 2.3.1 (Derivative). Let f be defined in a neighborhood of z_0 . The derivative of f at z_0 , denoted $f'(z_0)$, is:

$$f'(z_0) = \lim_{\Delta z \rightarrow 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$$

provided this limit exists.

If the limit exists, f is said to be **differentiable** at z_0 . The approach $\Delta z \rightarrow 0$ can occur along the real axis ($\Delta x \rightarrow 0$), the imaginary axis ($i\Delta y \rightarrow 0$), or any other path. The limit must be independent of the path.

Remark 2.3.1. Standard differentiation rules (sum, product, quotient, chain rule) from calculus apply to complex derivatives.

2.3.1 Examples of Differentiability

To illustrate the definition, let us look at one function that is differentiable everywhere and one that is not differentiable anywhere.

Example 2.3.1 (Differentiable Function). Consider the function $f(z) = z^2$. Using the definition of the derivative:

$$\begin{aligned} f'(z) &= \lim_{\Delta z \rightarrow 0} \frac{(z + \Delta z)^2 - z^2}{\Delta z} \\ &= \lim_{\Delta z \rightarrow 0} \frac{z^2 + 2z\Delta z + (\Delta z)^2 - z^2}{\Delta z} \\ &= \lim_{\Delta z \rightarrow 0} (2z + \Delta z) = 2z \end{aligned}$$

Since the limit $2z$ exists and is independent of the path taken by Δz , $f(z) = z^2$ is differentiable everywhere (it is an entire function).

Example 2.3.2 (Non-Differentiable Function). Consider the function $f(z) = \bar{z} = x - iy$. We attempt to find the limit of the difference quotient:

$$\lim_{\Delta z \rightarrow 0} \frac{\overline{z + \Delta z} - \bar{z}}{\Delta z} = \lim_{\Delta z \rightarrow 0} \frac{\overline{\Delta z}}{\Delta z}$$

Let $\Delta z = \Delta x + i\Delta y$.

- **Path 1 (Along real axis):** Set $\Delta y = 0$, so $\Delta z = \Delta x$.

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta x}{\Delta x} = 1$$

- **Path 2 (Along imaginary axis):** Set $\Delta x = 0$, so $\Delta z = i\Delta y$.

$$\lim_{\Delta y \rightarrow 0} \frac{-i\Delta y}{i\Delta y} = -1$$

Since the limits along these two paths are different ($1 \neq -1$), the derivative $f'(z)$ does not exist anywhere.

2.4 Cauchy-Riemann Equations

Since $f(z) = u(x, y) + iv(x, y)$, there is a necessary relationship between the partial derivatives of u and v for $f'(z)$ to exist.

Theorem 2.4.1 (Cauchy-Riemann Equations). *Suppose $f(z) = u(x, y) + iv(x, y)$ is differentiable at a point $z_0 = x_0 + iy_0$. Then the partial derivatives of u and v exist at (x_0, y_0) and satisfy the **Cauchy-Riemann equations**:*

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Furthermore, the derivative is given by:

$$f'(z_0) = u_x + iv_x$$

Proof. Assume $f(z) = u(x, y) + iv(x, y)$ is differentiable at a point z . Then the limit

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}$$

exists and is unique regardless of how Δz approaches 0.

Case 1: Approach along the horizontal axis ($\Delta y = 0$).

Let $\Delta z = \Delta x$. Then:

$$\begin{aligned} f'(z) &= \lim_{\Delta x \rightarrow 0} \frac{u(x + \Delta x, y) - u(x, y)}{\Delta x} + i \lim_{\Delta x \rightarrow 0} \frac{v(x + \Delta x, y) - v(x, y)}{\Delta x} \\ f'(z) &= u_x(x, y) + iv_x(x, y) \end{aligned} \quad (2.1)$$

Case 2: Approach along the vertical axis ($\Delta x = 0$).

Let $\Delta z = i\Delta y$. Then:

$$f'(z) = \lim_{\Delta y \rightarrow 0} \frac{u(x, y + \Delta y) - u(x, y)}{i\Delta y} + i \lim_{\Delta y \rightarrow 0} \frac{v(x, y + \Delta y) - v(x, y)}{i\Delta y}$$

Using $1/i = -i$, we get:

$$f'(z) = -iu_y(x, y) + v_y(x, y) \quad (2.2)$$

Equating the real and imaginary parts (2.1)-(2.2) gives the Cauchy-Riemann equations:

$$u_x = v_y \quad \text{and} \quad v_x = -u_y$$

□

Theorem 2.4.2 (Sufficient Conditions). *If the partial derivatives u_x, u_y, v_x, v_y exist, are continuous in a neighborhood of z_0 , and satisfy the Cauchy-Riemann equations at z_0 , then $f'(z_0)$ exists.*

2.4.1 Polar Form

Using $z = re^{i\theta}$, the Cauchy-Riemann equations in polar coordinates are:

$$ru_r = v_\theta, \quad u_\theta = -rv_r$$

and the derivative is given by:

$$f'(z) = e^{-i\theta}(u_r + iv_r)$$

Proof. Let $z = re^{i\theta}$ and $f(z) = u(r, \theta) + iv(r, \theta)$. Assuming f is differentiable at a non-zero point z , we can compute the partial derivatives with respect to r and θ using the chain rule.

Recall that $z = re^{i\theta}$.

1. Differentiate with respect to r :

$$\frac{\partial f}{\partial r} = f'(z) \frac{\partial z}{\partial r} = f'(z) e^{i\theta}$$

Also, directly from the components:

$$\frac{\partial f}{\partial r} = u_r + iv_r$$

Equating these implies:

$$f'(z) = e^{-i\theta}(u_r + iv_r) \quad (2.3)$$

2. Differentiate with respect to θ :

$$\frac{\partial f}{\partial \theta} = f'(z) \frac{\partial z}{\partial \theta} = f'(z)(ire^{i\theta}) = irf'(z)e^{i\theta}$$

Also, directly from the components:

$$\frac{\partial f}{\partial \theta} = u_\theta + iv_\theta$$

Solving for $f'(z)$:

$$f'(z) = \frac{1}{ire^{i\theta}}(u_\theta + iv_\theta) = \frac{e^{-i\theta}}{r}(v_\theta - iu_\theta) \quad (2.4)$$

Equating (2.3)-(2.4) gives:

$$u_r + iv_r = \frac{1}{r}v_\theta - i\frac{1}{r}u_\theta$$

Matching real and imaginary parts yields the polar form:

$$u_r = \frac{1}{r}v_\theta, \quad v_r = -\frac{1}{r}u_\theta$$

□

2.5 Analytic Functions

Definition 2.5.1 (Analyticity). A function f is **analytic** at a point z_0 if it is differentiable not only at z_0 but also at every point in some neighborhood of z_0 . A function is analytic in a domain D if it is analytic at every point in D .

Definition 2.5.2 (Entire Function). A function that is analytic at every point in the complex plane $\mathbb{C}$ is called an **entire function**. Examples include polynomials and e^z .

Theorem 2.5.1. If $f'(z) = 0$ everywhere in a domain D , then $f(z)$ is constant in D .

Definition 2.5.3 (Singular Point). A point z_0 is a **singular point** of a function f if f is not analytic at z_0 but is analytic at some point in every neighborhood of z_0 .

Example 2.5.1. The function $f(z) = \frac{1}{z}$ is analytic everywhere except at $z = 0$, which is a singular point.

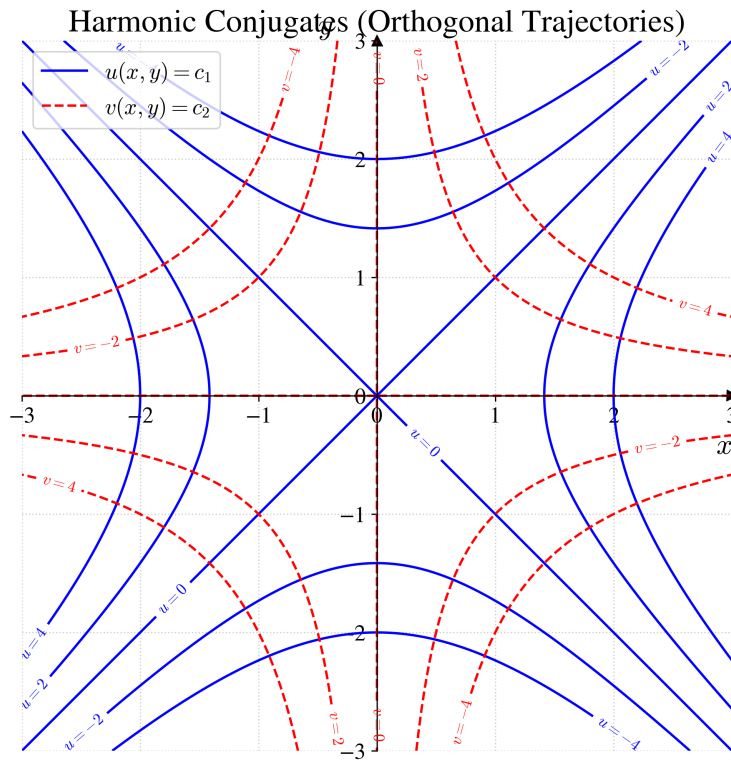


Figure 2.5: Harmonic functions and their conjugates.

2.6 Harmonic Functions

Analytic functions are closely related to the Laplace equation from physics and engineering.

Definition 2.6.1 (Harmonic Function). A real-valued function $H(x, y)$ is **harmonic** in a domain if it has continuous partial derivatives of the first and second order and satisfies Laplace's equation:

$$H_{xx}(x, y) + H_{yy}(x, y) = 0$$

Theorem 2.6.1. If $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain D , then its component functions u and v are harmonic in D .

Definition 2.6.2 (Harmonic Conjugate). If u and v are harmonic in a domain D and their first-order partial derivatives satisfy the Cauchy-Riemann equations, then v is called a **harmonic conjugate** of u .

2.7 Exercises

Exercise 2.7.1. Determine the domain of definition for the following functions:

1. $f(z) = \frac{1}{z^2+1}$
2. $f(z) = \text{Arg}\left(\frac{1}{z}\right)$

Exercise 2.7.2. Using the definition of the derivative (limits), show that the function $f(z) = \text{Re}(z)$ is nowhere differentiable.

Exercise 2.7.3. Use the Cauchy-Riemann equations to determine where the following functions are differentiable and where they are analytic:

1. $f(z) = \bar{z}$
2. $f(z) = e^{-x}(\cos y - i \sin y)$
3. $f(z) = z^2 + z$

Exercise 2.7.4. Show that the function $u(x, y) = 2x(1 - y)$ is harmonic. Find a function $v(x, y)$ such that $f(z) = u + iv$ is analytic. Express $f(z)$ in terms of z .

Exercise 2.7.5. Prove that if $f(z)$ is analytic in a domain D and $|f(z)|$ is constant in D , then $f(z)$ itself must be constant in D .

Exercise 2.7.6. Using the polar form of the Cauchy-Riemann equations, show that the function $f(z) = \sqrt{r}e^{i\theta/2}$ (where $r > 0$ and $-\pi < \theta < \pi$) is analytic in its domain. Compute $f'(z)$.

Exercise 2.7.7. Consider the function $f(z) = |z|^2$.

1. Write $f(z)$ in terms of x and y (i.e., find $u(x, y)$ and $v(x, y)$).
2. Apply the Cauchy-Riemann equations to determine the set of points where $f'(z)$ exists.
3. Is $f(z)$ analytic at any point? Explain the difference between your answer here and the result in part 2.

Chapter 3

Elementary Functions

3.1 The Exponential Function

We extend the real exponential function e^x to the complex plane in a way that preserves its fundamental properties (derivative, addition laws).

Definition 3.1.1 (Complex Exponential). Let $z = x + iy$. The exponential function e^z (or $\exp z$) is defined as:

$$e^z = e^x(\cos y + i \sin y)$$

This definition is motivated by Euler's formula $e^{iy} = \cos y + i \sin y$.

3.1.1 Properties

1. **Differentiation:** The derivative of e^z is e^z everywhere. Thus, e^z is an *entire* function.

2. **Modulus and Argument:**

$$|e^z| = e^x \quad \text{and} \quad \arg(e^z) = y + 2n\pi \quad (n \in \mathbb{Z})$$

3. **Non-Zero:** $e^z \neq 0$ for any complex number z .

4. **Periodicity:** The function is periodic with a purely imaginary period of $2\pi i$:

$$e^{z+2\pi i} = e^z$$

In view of the periodicity property, we define the **fundamental strip** for the exponential function as $\mathbb{R} \times [0, 2\pi)$, which maps bijectively to $\mathbb{C} \setminus \{0\}$ under e^z , see Figure 3.1

3.2 The Logarithmic Function

Now we must define the *inverse* of the complex exponential function, i.e. the **complex logarithm**.

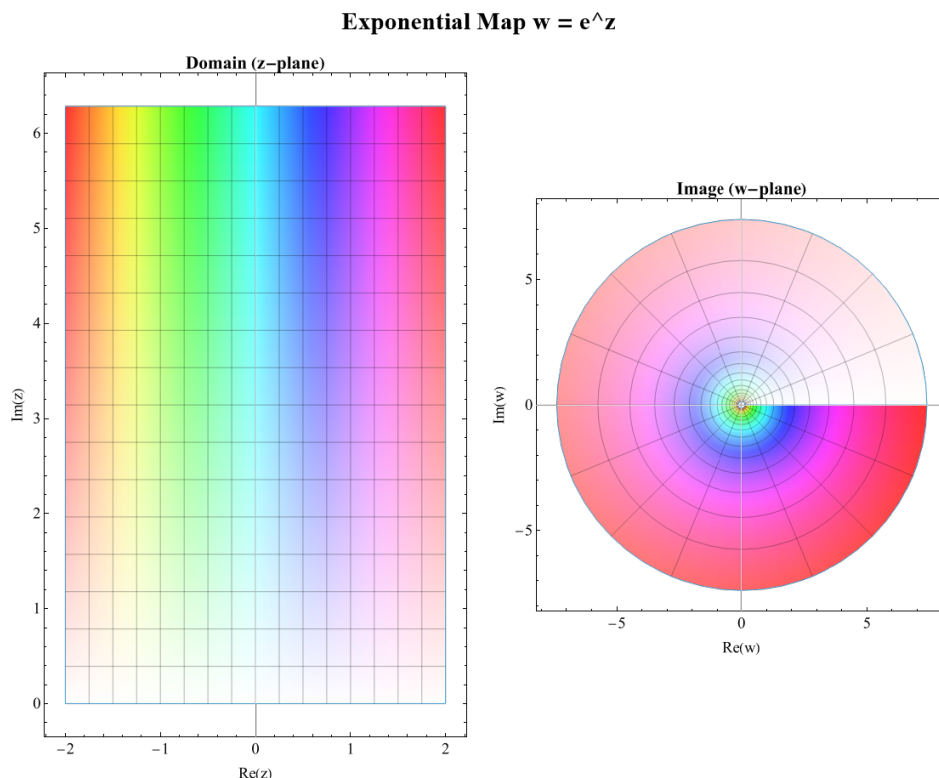


Figure 3.1: The mapping of the fundamental strip $\mathbb{R} \times [0, 2\pi)$ under the exponential function. Vertical lines map to circles, horizontal lines map to rays.

Definition 3.2.1 (Logarithm). For any nonzero complex number $z = re^{i\Theta}$ (where $\Theta = \text{Arg } z$), the logarithmic function is defined as:

$$\log z = \ln r + i(\Theta + 2n\pi) \quad (n \in \mathbb{Z})$$

or equivalently $\log z = \ln |z| + i \arg z$.

Definition 3.2.2 (Principal Value). The **Principal Value** of the logarithm, denoted by $\text{Log } z$, restricts the argument to the principal range $(-\pi, \pi]$:

$$\text{Log } z = \ln r + i\Theta \quad (-\pi < \Theta \leq \pi)$$

Remark 3.2.1. Notice that the $\text{Log}(z)$ is discontinuous along the negative real axis. Indeed for all $x \in \mathbb{R}, x > 0$ and $\varepsilon \in \mathbb{R}$

$$\lim_{\varepsilon \rightarrow 0^+} \text{Log}(-x + i\varepsilon) \neq \lim_{\varepsilon \rightarrow 0^+} \text{Log}(-x - i\varepsilon).$$

Exercise 3.2.1. Prove that for all $x > 0$

$$\lim_{\varepsilon \rightarrow 0^+} \text{Log}(-x + i\varepsilon) - \text{Log}(-x - i\varepsilon) = 2\pi i.$$

3.3 Branches and Derivatives of Logarithms

To treat the logarithm as a single-valued, differentiable function (suitable for calculus), we must restrict its domain.

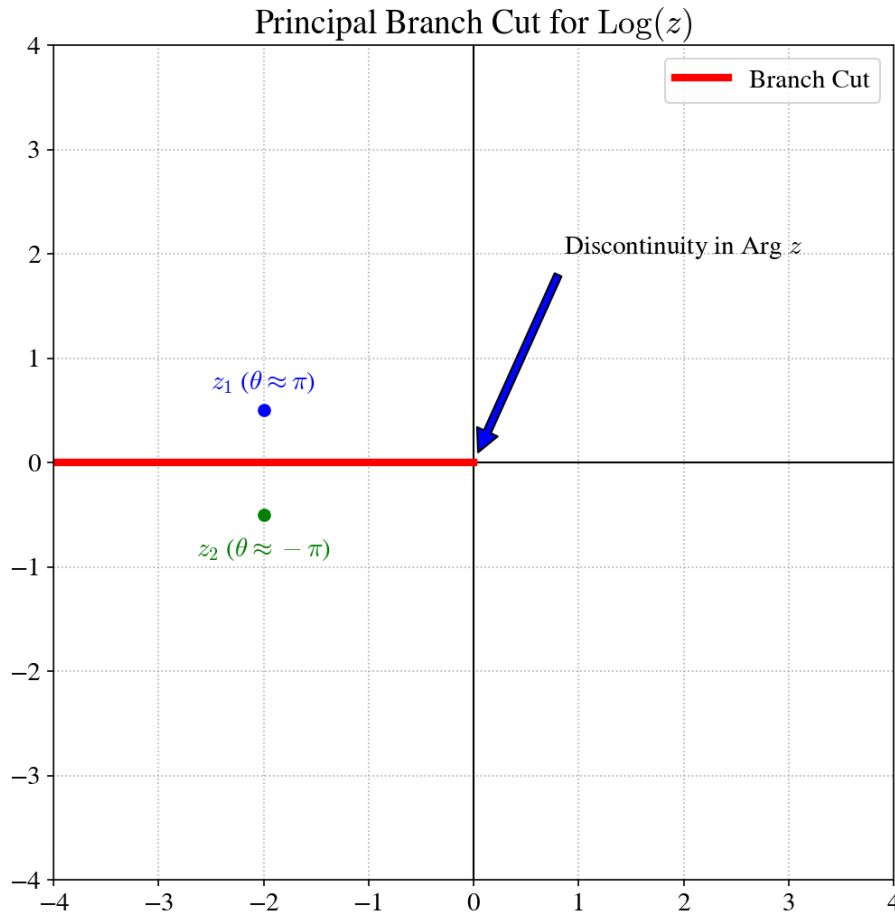


Figure 3.2: The branch-cut of the logarithm

Definition 3.3.1 (Branch of a Function). Let F be a single-valued function analytic in some domain D . If $F(z)$ is one of the values of a multiple-valued function $f(z)$ for every $z \in D$, then F is called a **branch** of f .

Definition 3.3.2 (Branch Cut). A **branch cut** is a portion of a line or curve that is introduced in order to define a branch F of a multiple-valued function f . Points on the branch cut are **singular points** (specifically, points of discontinuity) for that branch.

Example 3.3.1 (Principal Branch). The principal branch of the logarithm is:

$$\text{Log } z = \ln r + i\Theta \quad (r > 0, -\pi < \Theta < \pi)$$

Here, the branch cut is the **origin and the negative real axis** (where $\Theta = \pi$). The origin is called a **branch point**, see Fig 3.2

Theorem 3.3.1. For any branch of the logarithm defined on a domain D (not containing the origin), the derivative is:

$$\frac{d}{dz} \log z = \frac{1}{z}$$

3.3.1 Intermezzo: The Riemann Surface of the Logarithm

Introducing branch cuts is not the only way to make a multi-valued function single-valued. While a branch cut acts like a "brick wall" that stops us from walking completely around the origin and encountering multiple values, there is a much more elegant, deeply geometric approach: we can *change the space itself*.

Instead of restricting the function by cutting the plane, we can construct a new geometric domain called a **Riemann Surface**.

Imagine the flat complex plane (excluding the origin). As you walk around the origin in a circle, your angle θ continuously increases. On a normal, flat plane, after a full 360° (2π) rotation, you are standing on the exact same physical point where you started. This geometric collision is exactly what causes the logarithm to have multiple values at the same point!

But what if the complex plane wasn't flat? Imagine that as you walk counterclockwise around the origin, instead of returning to your starting point, you walk slightly upwards. The space behaves like an infinite spiral staircase or a circular parking garage ramp.

- After one full rotation (2π), you do not hit your starting point; you are on a new "sheet" directly above where you started.
- If you keep walking counterclockwise, you keep going up to higher and higher sheets.
- If you walk clockwise, your angle becomes negative, and you walk downwards onto lower sheets.

This infinite spiral is the Riemann surface for the complex logarithm.

Why is this so exciting? On this spiral surface, the point at angle 0 and the point at angle 2π are no longer the same point—they live on completely different levels of the staircase. Because every single continuous angle $\theta \in (-\infty, \infty)$ now has its own unique, physical location on the surface, the function $\log(z)$ only assigns **one value to one point**.

By allowing our 2D plane to stretch and warp into an infinite 3D spiral, we have completely "fixed" the logarithm! It becomes a perfectly smooth, single-valued, and continuous function everywhere on this new surface. This profound idea—solving algebraic problems (multi-valuedness) by inventing new geometric spaces—was introduced by Bernhard Riemann in 1851 and is the gateway to some of the most beautiful advanced mathematics, including Algebraic Geometry and Topology.

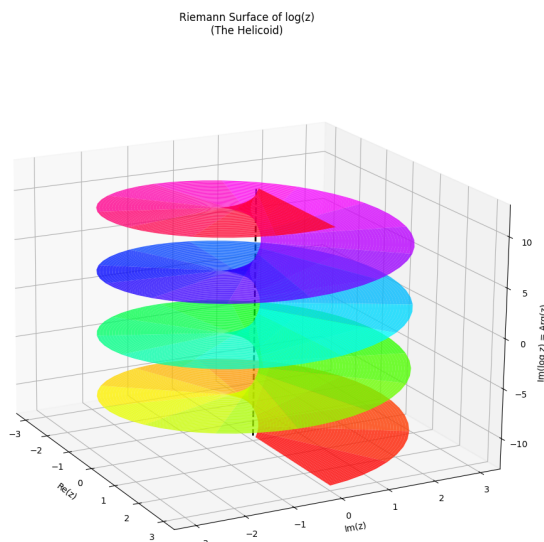
3.4 Trigonometric Functions

We define complex trigonometric functions using the exponential function, extending the Euler relations valid for real variables.

Definition 3.4.1 (Sine and Cosine).

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i}, \quad \cos z = \frac{e^{iz} + e^{-iz}}{2}$$

Other functions ($\tan z$, $\cot z$, $\sec z$, $\csc z$) are defined in terms of sine and cosine in the usual way.

Figure 3.3: The infinite spiral of the $\text{Log}(z)$

3.4.1 Properties

1. **Derivatives:**

$$\frac{d}{dz} \sin z = \cos z, \quad \frac{d}{dz} \cos z = -\sin z$$

2. **Identities:** The Pythagorean identity holds:

$$\sin^2 z + \cos^2 z = 1$$

3. **Zeros:**

- $\sin z = 0$ if and only if $z = n\pi$ (real zeros only).
- $\cos z = 0$ if and only if $z = \pi/2 + n\pi$ (real zeros only).

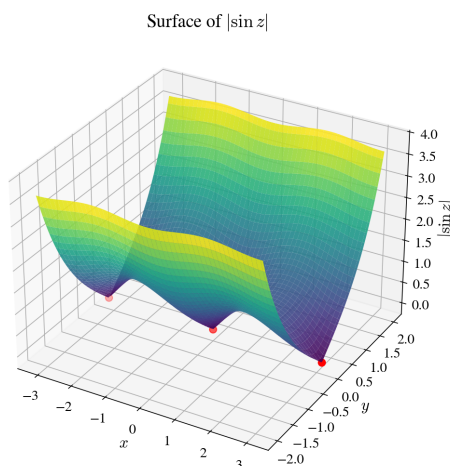
4. **Unboundedness:** Unlike real sine and cosine (bounded by 1), complex sine and cosine are **unbounded**. For example, on the imaginary axis $z = iy$:

$$|\sin(iy)| = \left| \frac{e^{-y} - e^y}{2i} \right| = \sinh y \rightarrow \infty \text{ as } y \rightarrow \infty$$

3.5 Hyperbolic Functions

Definition 3.5.1 (Sinh and Cosh). The hyperbolic sine and cosine are defined as:

$$\sinh z = \frac{e^z - e^{-z}}{2}, \quad \cosh z = \frac{e^z + e^{-z}}{2}$$

Figure 3.4: $|\sin(z)|$

3.5.1 Relations to Trigonometric Functions

The hyperbolic and trigonometric functions are intimately connected by the imaginary unit i :

$$\begin{aligned}\sinh(iz) &= i \sin z \\ \cosh(iz) &= \cos z \\ \sin(iz) &= i \sinh z \\ \cos(iz) &= \cosh z\end{aligned}$$

Theorem 3.5.1 (Identities).

$$\cosh^2 z - \sinh^2 z = 1$$

3.6 Complex Exponents

Now that we have defined the complex logarithm, we can extend the concept of exponents to complex numbers.

Definition 3.6.1 (Complex Powers). When $z \neq 0$ and c is any complex number, the function z^c is defined by means of the equation:

$$z^c = e^{c \log z}$$

Since $\log z$ is a multiple-valued function, the complex power z^c is also generally multiple-valued. The number of distinct values depends entirely on the nature of the exponent c .

Definition 3.6.2 (Principal Value of z^c). If $z \neq 0$, the **principal value** of z^c is obtained by using the principal branch of the logarithm:

$$z^c = e^{c \operatorname{Log} z}$$

3.6.1 Derivatives and Branch Cuts

Any branch of z^c is analytic where the chosen branch of $\log z$ is analytic. The branch cut for z^c will be identical to the branch cut of the logarithm used to define it. Using the chain rule, the derivative of a chosen branch of z^c is:

$$\frac{d}{dz} z^c = \frac{d}{dz} e^{c \log z} = e^{c \log z} \cdot \frac{c}{z} = cz^{c-1}$$

Example 3.6.1 (Computing i^{-i}). Let us find the principal value of i^{-i} . Using the definition:

$$i^{-i} = \exp(-i \operatorname{Log} i)$$

Since $|i| = 1$ and the principal argument is $\operatorname{Arg} i = \pi/2$, we have $\operatorname{Log} i = \ln(1) + i(\pi/2) = i\pi/2$. Therefore:

$$i^{-i} = \exp\left(-i \left(i \frac{\pi}{2}\right)\right) = \exp\left(\frac{\pi}{2}\right) = e^{\pi/2}$$

Notice that the principal value of i^{-i} is a purely real number!

3.6.2 A Deeper Look at Branch Cuts: Square Roots

It is tempting to manipulate complex exponents using familiar real-number algebra, such as writing $(ab)^c = a^c b^c$. However, because of branch cuts, this can fail spectacularly in the complex plane.

Let us analyze two functions that appear algebraically identical but have very different branch cuts when defined using the principal branch of the logarithm.

Function 1: $S(z) = (z^2 - 1)^{1/2}$ By definition, $S(z) = \exp\left(\frac{1}{2} \operatorname{Log}(z^2 - 1)\right)$. The principal logarithm has a branch cut whenever its argument is a negative real number or zero. Thus, $S(z)$ has a cut whenever $z^2 - 1 \in (-\infty, 0]$. Let $z = x + iy$. Then $z^2 - 1 = (x^2 - y^2 - 1) + i(2xy)$. For this to be real and non-positive, we need:

$$1. \quad 2xy = 0 \implies x = 0 \text{ or } y = 0.$$

$$2. \quad x^2 - y^2 - 1 \leq 0.$$

If $y = 0$ (the real axis), the condition becomes $x^2 - 1 \leq 0 \implies x \in [-1, 1]$.

If $x = 0$ (the imaginary axis), the condition becomes $-y^2 - 1 \leq 0$, which is true for *all* y . Therefore, the branch cut for $S(z)$ forms a cross: the line segment $[-1, 1]$ on the real axis, *and* the entire imaginary axis!

Function 2: $R(z) = (z-1)^{1/2}(z+1)^{1/2}$ By definition, $R(z) = \exp\left(\frac{1}{2} \operatorname{Log}(z-1)\right) \exp\left(\frac{1}{2} \operatorname{Log}(z+1)\right)$.

- The first term has a branch cut where $z-1 \in (-\infty, 0]$, which is the ray $x \leq 1, y = 0$.
- The second term has a branch cut where $z+1 \in (-\infty, 0]$, which is the ray $x \leq -1, y = 0$.

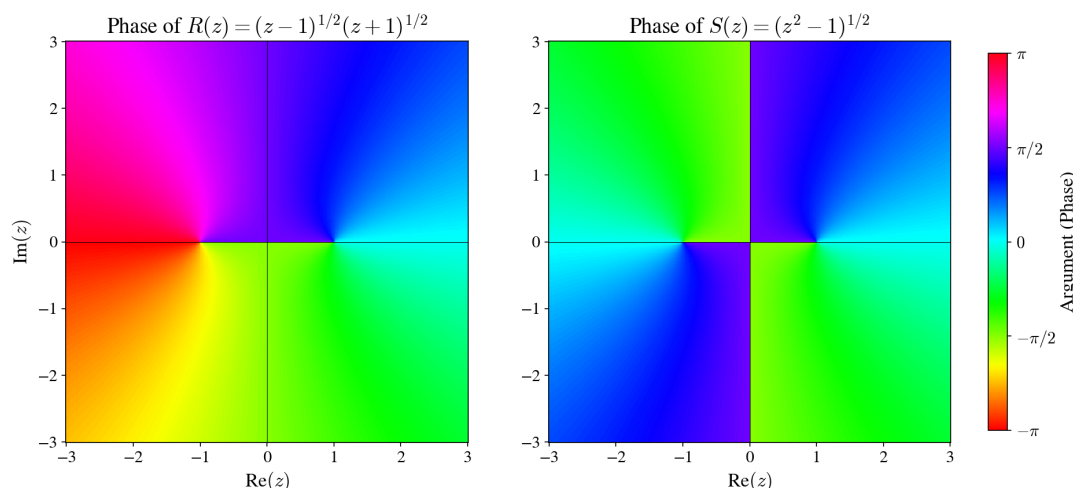


Figure 3.5: Comparison between the branch-cuts of $R(z) = (z-1)^{1/2}(z+1)^{1/2}$ and $S(z) = (z^2-1)^{1/2}$

It seems the cut for $R(z)$ is the entire ray $(-\infty, 1]$. However, let's look at the segment $(-\infty, -1)$ where both cuts overlap. If we approach the real axis from above ($y \rightarrow 0^+$), both $(z-1)$ and $(z+1)$ have an angle of π . The product evaluates to $e^{i\pi/2}e^{i\pi/2} = e^{i\pi} = -1$. If we approach from below ($y \rightarrow 0^-$), both terms have an angle of $-\pi$. The product evaluates to $e^{-i\pi/2}e^{-i\pi/2} = e^{-i\pi} = -1$. Because the limits from above and below agree, the discontinuity *cancels out*! Therefore, the branch cut for $R(z)$ is only the line segment $[-1, 1]$.

3.7 Exercises for Chapter 3

Exercise 3.7.1. Show that $\exp(z + \pi i) = -\exp z$.

Exercise 3.7.2. Find all values of z such that $e^z = -2$. (*Hint: Use the Logarithm*).

Exercise 3.7.3. Prove that $|\sin z|^2 = \sin^2 x + \sinh^2 y$. Use this to verify that $|\sin z|$ is unbounded.

Exercise 3.7.4. Compute the principal value $\text{Log}(-1 + i)$.

Exercise 3.7.5. Show that the function $f(z) = \text{Log}(z - i)$ has a branch point at $z = i$. Sketch a possible branch cut.

Exercise 3.7.6 (Values of Complex Powers). Using the definition $z^c = e^{c \log z}$, compute all possible values for the following complex powers. For each, explicitly identify the **principal value** (obtained by using the principal branch of the logarithm, $\text{Log}(z)$).

(a) $1^{\sqrt{2}}$

(b) $(-i)^i$

(c) $(1 + i)^{1-i}$

Exercise 3.7.7 (Branch Cuts of Complex Powers). Let $f(z) = (z + 2i)^i$.

- (a) Write the explicit definition of the *principal branch* of this function.
- (b) Determine the exact location of the branch cut for this principal branch. Which ray in the complex plane must be removed to make the function analytic? Sketch it.
- (c) Compute the derivative $f'(z)$ for this branch. Where is this derivative well-defined?

Exercise 3.7.8 (Analyzing Branch Cuts of Algebraic Functions). Consider the function $g(z)$ defined using the principal branch of the complex logarithm:

$$g(z) = \exp\left(\frac{1}{2}\text{Log}(z^2 + 1)\right)$$

- (a) For what values of w does $\text{Log}(w)$ have its branch cut?
- (b) Set $w = z^2 + 1$. By letting $z = x + iy$, find the exact set of points (x, y) in the complex plane where $z^2 + 1$ falls on the branch cut of the principal logarithm.
- (c) Sketch the resulting branch cuts for $g(z)$.

Hint: The cuts should lie on the imaginary axis. Find their exact endpoints.

Exercise 3.7.9 (Overlapping and Canceling Branch Cuts). Consider the multi-valued function $h(z) = z^{1/2}(z - 2)^{1/2}$. Suppose we define a specific branch of $h(z)$ by taking the principal branch of the square root for both terms:

$$H(z) = \exp\left(\frac{1}{2}\text{Log}(z)\right) \exp\left(\frac{1}{2}\text{Log}(z - 2)\right)$$

- (a) Identify the individual branch cuts for $\exp\left(\frac{1}{2}\text{Log}(z)\right)$ and $\exp\left(\frac{1}{2}\text{Log}(z - 2)\right)$.
- (b) Consider the segment of the negative real axis $(-\infty, 0)$. This segment lies on the branch cut for *both* terms. Pick a point $x_0 < 0$ on this segment. Calculate the limit of $H(z)$ as $z \rightarrow x_0$ from the upper half-plane ($y \rightarrow 0^+$) and from the lower half-plane ($y \rightarrow 0^-$).
- (c) Show that these two limits are exactly the same. Conclude that the branch cut "cancels out" on $(-\infty, 0)$ and that the true branch cut for $H(z)$ is strictly the line segment $[0, 2]$.